SUMS OF TWO SQUARES WITH FEW PRIME FACTORS, II: THE CUSPIDAL ERROR IN MEAN SQUARE AND ALMOST ALL N

RUQING CHEN

ABSTRACT. Let m, m' be coprime odd integers, $\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z)$, $G = G_{m,m'}$ its cuspidal part and $L = 4mm'$, so that $a_G(N) = T(m, m'; N) - M(m, m'; N)$ is the error term in the number of representations of N by $m(x^2 + y^2) + m'(z^2 + w^2)$. We prove the mean-square bound $\sum_{N\le Y} |a_G(N)|^2 \le 4\pi e^2\, Y(3 + 107Y/L)\|G\|^2$, which with Waibel's norm bound gives $\sum_{N\le Y} |T-M|^2 \ll Y^2(mm')^{-1/2+\varepsilon}$ for $Y \ge mm'$: on average over N the error is $N^{1/2}(mm')^{-1/4}$, the size observed numerically in Part I. It follows that the weighted sequence $r(n)r(N-n)$ has level of distribution $2/3$ for all but $O(X^{1-\delta})$ integers $N \equiv 2 \pmod 4$ up to X, and for these N the sieve of Part I gives $\gg N/(\log N)^2$ integers $n < N$ with n, $N-n$ sums of two squares, $\Omega(n(N-n)) \le 6$ and $\max(\Omega(n), \Omega(N-n)) \le 4$. We reformulate the uniform conjecture of Part I as a near-orthogonality statement between $\Theta_{m,m'}$ and the Poincaré series, show that it holds on average in a strong form, and record numerical evidence that $\|G_{m,m'}\|^2 \approx 3\,(mm')^{1/4}$, smaller than the general bound.

1. RESULTS

We keep the notation of [1], referred to as Part I. Thus $r(n)$ is the number of representations of n as a sum of two squares, m, m' are coprime odd positive integers,

$$T(m, m'; N) = \sum_{ma+m'b=N} r(a)r(b), \qquad L = 4mm',$$

$$\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z) = \sum_{N\ge 0} T(m, m'; N)e(Nz) \in M_2(\Gamma_0(L)),$$

$\Theta_{m,m'} = E + G$ is the decomposition into the Eisenstein part, whose coefficients are the Siegel main terms $M(m, m'; N)$, and the cuspidal part $G = G_{m,m'} \in S_2(\Gamma_0(L))$, so that

$$a_G(N) = T(m, m'; N) - M(m, m'; N) \qquad \text{for every } N \ge 1.$$

The Petersson norm is $\|G\|^2 = \int_{\Gamma_0(L)\backslash\mathbb{H}} |G|^2 \,\mathrm{d}x\,\mathrm{d}y$, and Lemma 2.1 of Part I (Waibel's theorem) gives $\|G\| \ll (mm')^{1/4+\varepsilon}$. Part I proves $a_G(N) \ll d(N)N^{1/2}(mm')^{1/4+\varepsilon}$ for $(N, mm') = 1$, $v_2(N) \le 1$, and conjectures (Conjecture 5.1 there) that the factor $(mm')^{1/4}$ can be removed. The first result of this note is that the conjecture holds in mean square over N, with room to spare.

Theorem 1.1 (Mean square)**.** *Let $L \ge 1$, $G \in S_2(\Gamma_0(L))$ and $Y \ge 1$. Then*

$$\sum_{n\le Y} |a_G(n)|^2 \;\le\; 4\pi e^2\, Y\Big(3 + \frac{107\,Y}{L}\Big)\|G\|^2. \tag{1}$$

In particular, for coprime odd m, m', $G = G_{m,m'}$ and $Y \ge mm'$,

$$\sum_{N\le Y} \big|T(m, m'; N) - M(m, m'; N)\big|^2 \;\ll_\varepsilon\; Y^2\,(mm')^{-1/2+\varepsilon}, \tag{2}$$

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and hence, since $\sum_{Y/2<N\le Y}|a_G(N)|^2/N \le (2/Y)\sum_{N\le Y}|a_G(N)|^2$, *the root mean square of* $(T-M)/\sqrt{N}$ *over* $Y/2<N\le Y$ *is* $\ll (mm')^{-1/4+\varepsilon}$.

The bound (2) is what the data of Part I, Table 3, show (the root mean square there decreases with mm'; see Section 5 for a closer look). The proof of Theorem 1.1 is a classical covering argument and does not use the Petersson formula; the dependence on the level is exactly the factor Y/L, i.e. the volume of $\Gamma_0(L)\backslash\mathbb{H}$.

Averaging over N is natural for the arithmetic application, since the sieve of Part I is applied to one N at a time and an exceptional set of N can be tolerated. Let, as in Part I, $N\equiv 2 \pmod 4$, $\mathcal{A}=\{n\le N:\ (n,N)=1\}$, $w(n)=r(n)r(N-n)$, and for coprime square-free d,e with $(de,2N)=1$ let $R(d,e)=R(d,e;N)$ be the remainder in

$$\sum_{\substack{n\in\mathcal{A}\\ d\mid n,\ e\mid N-n}} w(n) = X_N\, h(d)h(e) + R(d,e;N)$$

(this is $S(d,e)=M(d,e)+R(d,e)$ of Part I, Section 3; we write X_N for the main term X there, to avoid a clash with the length X of the range of N below).

Theorem 1.2 (Level of distribution 2/3 for almost all N)**.** *Let* $0<\varepsilon<1/20$, $X\ge X_0(\varepsilon)$ *and* $D=X^{2/3-2\varepsilon}$. *Then*

$$\sum_{\substack{X/2<N\le X\\ N\equiv 2\ (4)}}\ \sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \ll_\varepsilon\ X^{2-\varepsilon/2}.$$

Consequently, for all but $O_\varepsilon(X^{1-\varepsilon/4})$ *integers* $N\equiv 2 \pmod 4$ *in* $(X/2,X]$,

$$\sum_{\substack{de\le N^{2/3-2\varepsilon}\\ (d,e)=1,\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \le\ N^{1-\varepsilon/5}. \tag{3}$$

For comparison, Theorem 3.2 of Part I gives (3) with 2/5 in place of 2/3 for every N, and Conjecture 5.1 of Part I would give 1/2 for every N. The sieve of Part I, Section 4, uses the level of distribution only through (3); running it at $\theta=2/3$ gives the following.

Corollary 1.3. *For all but* $O(X^{1-\delta})$ *integers* $N\equiv 2 \pmod 4$ *in* $(X/2,X]$, *with an absolute* $\delta>0$,

$$\#\big\{1\le n<N:\ n=x^2+y^2,\ N-n=z^2+w^2,\ \Omega(n(N-n))\le 6,\ \max(\Omega(n),\Omega(N-n))\le 4\big\}$$
$$\ge\ 0.13\,\mathfrak{S}_r(N)\,\frac{N}{(\log N)^2},\qquad \mathfrak{S}_r(N)=\prod_{\substack{p\mid N\\ p>2}}\frac{p-1}{p-2},$$

the integers counted being coprime to N *and products of distinct primes* $\equiv 1 \pmod 4$ *exceeding* $N^{3/50}$.

The pair $(6,4)$ should be compared with $(9,6)$ for every N (Part I, Theorem 1.1) and $(8,5)$ for every N under Conjecture 5.1 of Part I. The certified sieve parameters are $\alpha=3/50$, $\lambda=1/3$, $\beta_1=51/100$ (Section 3).

Section 4 returns to the uniform conjecture. Writing P_N for the N-th Poincaré series of $S_2(\Gamma_0(L))$, one has $a_G(N)=4\pi N\langle G,P_N\rangle$, so that

$$|a_G(N)|^2 = 4\pi N\,\Delta_L(N,N)\,\|G\|^2\cos^2\angle(G,P_N),$$

where $\Delta_L(N,N)$ is the Petersson density of Part I, (11). Theorem 1.1 says that the average over $N\le Y$ of $\Delta_L(N,N)\cos^2\angle(G,P_N)$ is $O(1/L)$ for $Y\ge L$, the value one would get from

a random vector in the $\asymp L$-dimensional space $S_2(\Gamma_0(L))$; Conjecture 5.1 of Part I asks only that no single N exceed this average by more than a factor $N^{\varepsilon}L/\|G\|^2 \approx (mm')^{1/2}$. Section 5 contains the numerical evidence for Theorem 1.1 and the estimate $\|G_{m,m'}\|^2 \approx 3(mm')^{1/4}$ from the Rankin–Selberg asymptotic.

2. Proof of Theorem 1.1

Let $G(z) = \sum_{n\ge1} a(n)e(nz) \in S_2(\Gamma_0(L))$ and put $F(z) = y^2|G(z)|^2$, $z = x+iy$. Since $G(\gamma z) = (cz+d)^2 G(z)$ and $\operatorname{Im}(\gamma z) = y/|cz+d|^2$ for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \Gamma_0(L)$, the function F is $\Gamma_0(L)$-invariant, and with the invariant measure $\mathrm{d}\mu = y^{-2}\mathrm{d}x\,\mathrm{d}y$ we have $\|G\|^2 = \int_{\mathcal F} F\,\mathrm{d}\mu$ for any fundamental domain $\mathcal F$ of $\Gamma_0(L)$.

Let $y_0 = 1/(4\pi Y)$ and $R = \{x+iy :\ 0\le x\le 1,\ y_0 \le y \le 2y_0\}$. By Parseval, $\int_0^1 |G(x+iy)|^2\,\mathrm{d}x = \sum_n |a(n)|^2 e^{-4\pi n y}$, hence

$$\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \sum_n |a(n)|^2 \int_{y_0}^{2y_0} e^{-4\pi n y}\,\mathrm{d}y \ \ge\ \frac{e^{-2}}{4\pi Y}\sum_{n\le Y}|a(n)|^2,$$

because $4\pi n y \le 8\pi Y y_0 = 2$ for $n\le Y$, $y\le 2y_0$. On the other hand $\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \int_R F\,\mathrm{d}\mu$, and

$$\int_R F\,\mathrm{d}\mu = \sum_{\gamma\in\Gamma_0(L)/\{\pm1\}} \int_{R\cap\gamma\mathcal F} F\,\mathrm{d}\mu = \int_{\mathcal F} F(w)\,\#\{\gamma :\ \gamma w\in R\}\,\mathrm{d}\mu(w) \ \le\ \mu_R\,\|G\|^2,$$

where $\mu_R = \sup_{z\in R}\#\{\gamma\in\Gamma_0(L)/\{\pm1\} :\ \gamma z\in R\}$ (the two suprema, over $w\in\mathcal F$ and over $z\in R$, coincide, since $\#\{\gamma : \gamma w\in R\} = \#\{\gamma : \gamma z\in R\}$ for $z = \gamma_0 w$). It remains to bound μ_R. Fix $z = x+iy\in R$ and let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \Gamma_0(L)$ with $\gamma z\in R$.

- If $c = 0$, then $\gamma = \pm\left(\begin{smallmatrix} 1 & b \\ 0 & 1\end{smallmatrix}\right)$ and $\gamma z = z+b$, so $b\in\{-1,0,1\}$: at most 3 elements.
- If $c\ne 0$, we may take $c>0$, since γ and $-\gamma$ are the same element of $\Gamma_0(L)/\{\pm1\}$. Then $L\mid c$, and $\operatorname{Im}(\gamma z) = y/|cz+d|^2 \in [y_0,2y_0]$ together with $y\in[y_0,2y_0]$ gives $|cz+d|^2\in[1/2,2]$. Since $|cz+d|^2 = (cx+d)^2 + c^2y^2$, this forces $c^2y^2\le 2$, hence $c\le \sqrt{2}/y_0 = 4\sqrt{2}\,\pi Y$, and $|cx+d|\le\sqrt{2}$. The number of positive multiples c of L with $c\le 4\sqrt{2}\pi Y$ is at most $4\sqrt{2}\pi\,Y/L$; for each c there are at most 3 integers d with $|cx+d|\le\sqrt{2}$; and for each pair (c,d) the elements of $\Gamma_0(L)$ with bottom row (c,d) are $\left(\begin{smallmatrix} 1 & k \\ 0 & 1\end{smallmatrix}\right)\gamma_0$, $k\in\mathbb Z$, whose images $\gamma_0 z + k$ have real part in $[0,1]$ for at most 2 values of k. This gives at most $4\sqrt{2}\pi\cdot 3\cdot 2\,Y/L = 24\sqrt{2}\pi\,Y/L < 107\,Y/L$ elements.

Hence $\mu_R \le 3 + 107Y/L$, and

$$\sum_{n\le Y}|a(n)|^2 \le 4\pi e^2 Y\int_R F\,\mathrm{d}\mu \le 4\pi e^2\,Y\Big(3+\frac{107Y}{L}\Big)\|G\|^2,$$

which is (1). For $G = G_{m,m'}$, $L = 4mm'$ and $Y\ge mm'$ the bracket is $\le 3+107Y/(4mm') \le 30Y/(mm')$, and Lemma 2.1 of Part I gives (2). □

Remark 2.1. The argument is the classical one for bounding $\sum_{n\le Y}|a(n)|^2$ by the norm (compare [2, Chapter 5]); its only input is the invariance of $y^2|G|^2$. The strip $y_0\le y\le 2y_0$ with $y_0 = 1/(4\pi Y)$ is chosen so that the exponential factor in Parseval's identity stays at e^{-2} for all $n\le Y$; the strip $1/Y\le y\le 2/Y$ would give the same argument with $e^{8\pi}$ in place of $4\pi e^2$. For $Y\ge L$ the factor Y^2/L in (1) is of the correct order of magnitude: by the Rankin–Selberg method, $\sum_{n\le Y}|a(n)|^2/n \sim 16\pi^2\|G\|^2 Y/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb H)$ with $\operatorname{vol} = \frac{\pi}{3}L\prod_{p\mid L}(1+1/p)$ (measure $\mathrm{d}\mu$), so that $\sum_{n\le Y}|a(n)|^2 \sim 8\pi^2\|G\|^2Y^2/\operatorname{vol}$ for fixed G as $Y\to\infty$. Theorem 1.1 makes this uniform in the level, at the cost of the constant.

3. Proof of Theorem 1.2 and Corollary 1.3

3.1. **The remainder on average.** We follow the proof of Theorem 3.2 of Part I, replacing the uniform bound for $|T-M|$ by Theorem 1.1 after an application of Cauchy–Schwarz over N. Recall from Part I, (15), that for $N\equiv 2 \pmod 4$ and coprime square-free d,e with $(de,2N)=1$,

$$R(d,e;N)=\sum_{t\mid N^\flat}\mu(t)\sum_{m\mid d^2}\sum_{m'\mid e^2}c_d(m)c_e(m')\sum_{\substack{t_1\mid t^2\\ t\mid t_1}}c_t(t_1)\big(a_G(N)-a_G(N/2)\big),\qquad G=G_{t_1m,m'},\tag{4}$$

where $N^\flat$ is the product of the odd primes dividing N, $m=ds$, $m'=es'$ with $s\mid d$, $s'\mid e$, and likewise $t_1=tu$ with $u\mid t$ (the coefficients $c_d(m)$, $c_t(t_1)$ of Part I, (14), vanish unless $m=ds$ with $s\mid d$, resp. $t_1=tu$ with $u\mid t$; in particular $t\mid t_1$); the coefficients t_1m,m' are coprime, odd and cube-free, $\tilde m\tilde m':=t_1mm'=t_1ss'\cdot de$, and the coefficients c satisfy $|c_d(m)c_e(m')c_t(t_1)|\le 3^{\omega(d)+\omega(e)+\omega(t)}$. Let $\eta=\varepsilon/10$. For $N\le X$ with $X\ge X_0(\varepsilon)$ we have, as in (F1)–(F3) of Part I: $d_3(de)\,3^{\omega(d)+\omega(e)+\omega(t)}\,4^{\omega(N)}d(N)^2\le X^\eta$; for fixed (d,e,N) the number of tuples (s,s',t,t_1) is at most X^η; and $\sum_{k\le D}d(k)/k\le X^\eta$, $\sum_{k\le D}d(k)k^{-1/4}\le D^{3/4}X^\eta$. We also use the two bounds for a single term,

$$|a_G(N')|\le T(\tilde m,\tilde m';N')+M(\tilde m,\tilde m';N')\ll X^{2\eta}\Big(\frac{X}{\tilde m\tilde m'}+1\Big)\qquad (N'\le X),\tag{5}$$

$$\sum_{n\le X}|a_G(n)|\le X^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}\ll X\Big(1+\frac{X}{L}\Big)^{1/2}L^{1/4+\eta},\tag{6}$$

where $G=G_{\tilde m,\tilde m'}$ and $L=4\tilde m\tilde m'$, the first being (20) of Part I, the second Cauchy–Schwarz, (1) and Lemma 2.1 of Part I; for $L\le X$ the right-hand side of (6) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/4+\eta}$. Let Σ denote the double sum in Theorem 1.2. Inserting (4) and splitting the tuples according to the size of t_1ss':

Tuples with $t_1ss'>X^\varepsilon$. For each N and each tuple, (5) with $\tilde m\tilde m'>X^\varepsilon de$ gives $|a_G(N')|\ll X^{2\eta}(X^{1-\varepsilon}/(de)+1)$. Summing over the $\le X^\eta$ tuples, over $N\le X$ and over $de\le D$ with the weights $d_3(de)\cdot 3^\omega\le X^\eta$, these tuples contribute

$$\ll X^{1+4\eta}\sum_{k\le D}d(k)\Big(\frac{X^{1-\varepsilon}}{k}+1\Big)\ll X^{2-\varepsilon+5\eta}+X^{1+5\eta}D\ll X^{2-\varepsilon/2}.$$

Tuples with $t_1ss'\le X^\varepsilon$. Now $t\le t_1\le X^\varepsilon$ (as $t\mid t_1$), so t runs over at most X^ε values, t_1 over at most $d(t^2)\le X^\eta$ values, and (s,s') over at most $d(d)d(e)\le X^\eta$ values; the forms $G=G_{t_1m,m'}$ have level $L=4t_1ss'de\le 4X^\varepsilon D\le X$, so that the right-hand side of (6) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/4+\eta}$. For a fixed tuple and fixed (d,e), we drop the condition $t\mid N$ except for the count: the number of $N\le X$ with $t\mid N$ is $\le X/t$, and Cauchy–Schwarz gives

$$\begin{aligned}\sum_{\substack{N\le X\\ t\mid N}}\big(|a_G(N)|+|a_G(N/2)|\big)&\le 2\Big(\frac{X}{t}\Big)^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}\\&\ll\Big(\frac{X}{t}\Big)^{1/2}X\,(\tilde m\tilde m')^{-1/4+\eta}\le X^{3/2+\eta}\,t^{-1/2}(de)^{-1/4},\end{aligned}$$

using (1) in the form $\sum_{n\le X}|a_G(n)|^2\ll X^2\|G\|^2/L$ and $(\tilde m\tilde m')^{-1/4+\eta}\le X^\eta(de)^{-1/4}$. Summing over $t\le X^\varepsilon$ ($\sum_t t^{-1/2}\le 2X^{\varepsilon/2}$), over the other tuple variables ($\le X^{2\eta}$), and over $de\le D$ with the weights $\le X^\eta$,

$$\ll X^{3/2+4\eta+\varepsilon/2}\sum_{k\le D}d(k)k^{-1/4}\ll X^{3/2+5\eta+\varepsilon/2}D^{3/4}=X^{2-\varepsilon+5\eta}=X^{2-\varepsilon/2},$$

since $D^{3/4} = X^{1/2-3\varepsilon/2}$. Both contributions are $\ll X^{2-\varepsilon/2}$, which proves the first assertion.

For the second, let $S(N)$ be the inner sum with $D = X^{2/3-2\varepsilon}$. The number of $N \in (X/2, X]$, $N \equiv 2 \pmod 4$, with $S(N) > X^{1-\varepsilon/4}$ is at most $\Sigma/X^{1-\varepsilon/4} \ll X^{1-\varepsilon/4}$. For the remaining N, since $N^{2/3-2\varepsilon} \le D$ and $X^{1-\varepsilon/4} \le (2N)^{1-\varepsilon/4} \le N^{1-\varepsilon/5}$ for large X, (3) holds. □

Remark 3.1. The exponent $2/3$ is the balance $X^{3/2}D^{3/4} = X^2$ between the mean-square bound $X^{3/2}(de)^{-1/4}$ for the sum over N of one remainder and the number D of moduli, exactly as $2/5$ in Part I is the balance $N^{1/2}D^{1/4} \cdot D = N$ for a single N. Any improvement of the exponent $1/4$ in Waibel's bound, say $\|G_{m,m'}\| \ll (mm')^{\beta+\varepsilon}$, gives the level $1/(2(1+\beta))$ for every N (Part I, Remark 3.3) and, through the balance $X^{3/2}D^{1/2+\beta} = X^2$, the level $1/(1+2\beta)$ for almost all N; for $\beta = 1/4$ these are $2/5$ and $2/3$, and for $\beta = 1/8$ (Section 5) they would be $4/9$ and $4/5$.

Remark 3.2. The exceptional set $X^{1-\varepsilon/4}$ comes from Markov's inequality applied to a first moment of $S(N)$, which in turn rests on the second moment of $a_G(N)$. A bound for a higher moment, $\sum_{N\le Y} |a_G(N)|^{2k} \ll Y^{k+1}(mm')^{-k/2+\varepsilon}$, would shrink the exceptional set to $X^{1-k\varepsilon'}$; we have no such bound for $k \ge 2$, and do not pursue this.

3.2. **The sieve at level** $2/3$**.** Proposition 4.3 of Part I and its proof use the level of distribution only through the bound (3) (there with $2/5$); they hold verbatim, for a given N, with $\theta = 2/5$ replaced by any $\theta \in (0,1)$ for which (3) holds with $N^{\theta-\varepsilon}$ in place of $N^{2/3-2\varepsilon}$, provided $\alpha < \beta_1 < \theta$. Lemmas 4.1 and 4.2 of Part I do not involve the level at all. Hence, for every N satisfying (3), the conclusion of Proposition 4.3 holds with $\theta = 2/3$ (the ε there being 2ε here), and the proof of Theorem 1.1 of Part I goes through with the parameters certified for $\theta = 2/3$. The script `certify_modular.py` of Part I, extended by the argument `2/3` in the repository of this paper, certifies in interval arithmetic

$$\theta = \tfrac{2}{3}, \qquad \alpha = \tfrac{3}{50}, \qquad \lambda = \tfrac{1}{3}, \qquad \beta_1 = \tfrac{51}{100},$$
$$\tfrac{1}{\lambda} + \tfrac{2}{\beta_1} = \tfrac{353}{51} < 7, \qquad \tfrac{1}{\lambda} + \tfrac{1}{\beta_1} = \tfrac{253}{51} < 5, \qquad L - \lambda R \ge 0.036741$$

(with $L \ge 0.998974$, $R \le 2.886700$, $\theta_A = 0.3330$; a second set $\alpha = 29/500$, $\beta_1 = 13/25$ gives $L - \lambda R \ge 0.012502$; the output files are listed at the end of the paper). By Lemma 4.2 of Part I this gives $\Omega(n(N-n)) \le 6$ and $\max(\Omega(n), \Omega(N-n)) \le 4$, and the weight of each n counted is $w(n) = 16 \cdot 2^{\Omega(n(N-n))} \le 2^{10}$. As in the proof of Theorem 1.1 of Part I, the number of integers counted is at least

$$\big(0.036741 + o(1)\big)\, 2^{-10} \cdot 8\, \mathfrak{S}_r(N)\, c_{2h}\, \frac{N}{\alpha^2 (\log N)^2} \;\ge\; 0.13\, \mathfrak{S}_r(N) \frac{N}{(\log N)^2},$$

since $0.036741 \cdot 2^{-10} \cdot 8 \cdot 1.66485 \cdot (50/3)^2 = 0.1327\ldots$. The exceptional set is that of Theorem 1.2 with, say, $\varepsilon = 1/40$, so $\delta = 1/160$ is admissible. □

Remark 3.3. At the level $1/2$ the same sieve gives $(8,5)$ (Part I, Remark 4.5), at $2/3$ it gives $(6,4)$. The floating-point scan of `modular_sieve_scan.py` at $\theta = 2/3$ shows that $(6,4)$ is the limit of the method at this level: the best value of $1/\lambda + 2/\beta_1$ is about 6.8 and that of $1/\lambda + 1/\beta_1$ about 4.65, so neither $(5,\cdot)$ nor $(\cdot,3)$ is reachable with these weights. Corollary 1.3 is an "almost all N" statement with a power-saving exceptional set; it says nothing about any individual N, and in particular nothing about the integers N of Theorem 1.1 of Part I beyond what is proved there.

4. The uniform conjecture as a statement about Poincaré series

Let $L = 4mm'$ and let $P_N \in S_2(\Gamma_0(L))$, $N \geq 1$, be the Riesz representative of the N-th coefficient: the unique cusp form with

$$\langle f, P_N \rangle = \frac{a_f(N)}{4\pi N} \qquad \text{for all } f \in S_2(\Gamma_0(L)), \tag{7}$$

i.e. $P_N = (4\pi N)^{-1} \sum_{f \in \mathcal{B}} \overline{a_f(N)}\, f$ for an orthonormal basis $\mathcal{B}$. (This is the N-th Poincaré series; for weight 2 the series itself converges only conditionally, which is why we define P_N by (7).) By (11) of Part I,

$$\|P_N\|^2 = \frac{a_{P_N}(N)}{4\pi N} = \frac{1}{(4\pi N)^2} \sum_{f \in \mathcal{B}} |a_f(N)|^2 = \frac{\Delta_L(N,N)}{4\pi N}.$$

Hence, for $G = G_{m,m'}$,

$$|a_G(N)|^2 = (4\pi N)^2 |\langle G, P_N \rangle|^2 = 4\pi N\, \Delta_L(N,N)\, \|G\|^2 \cos^2 \angle(G, P_N), \tag{8}$$

where $\cos \angle(G, P_N) = |\langle G, P_N \rangle| / (\|G\|\, \|P_N\|)$. This identity separates the three quantities that enter any bound for $a_G(N)$.

(i) $\|G\|^2$: controlled by Waibel's theorem, $\ll (mm')^{1/2+\varepsilon}$; numerically $\approx 3(mm')^{1/4}$ (Section 5).
(ii) $\Delta_L(N,N)$: the Petersson density; $\leq C'$ for $N \leq L^{1/2}$ (Part I, Lemma 2.2, Step 4), and $1 + O(N^{1/2+\varepsilon}/L)$ in general, which is all that is known for $N \gg L$.
(iii) $\cos^2 \angle(G, P_N)$: the alignment of the theta series with the Poincaré series. Lemma 2.2 of Part I bounds it by 1 (after moving the coprime part of N into a Hecke operator, which replaces P_N by P_{n_2} with $n_2 \in \{1, 2\}$ and G by $T_{n_1} G$).

Theorem 1.1 is a statement about (iii) on average. Inserting (8) into (1) and restricting to $N \in (Y/2, Y]$,

$$\frac{1}{Y/2} \sum_{Y/2 < N \leq Y} \Delta_L(N,N) \cos^2 \angle(G, P_N) \;\leq\; 4e^2 \Big(\frac{3}{Y} + \frac{107}{L} \Big) \;\ll\; \frac{1}{L} \qquad (Y \geq L). \tag{9}$$

Since $\dim S_2(\Gamma_0(L)) \asymp L$, the right-hand side is the value of $\cos^2$ for a unit vector in general position: on average over N, the theta series is no more aligned with the Poincaré series than a random cusp form of level L would be. In these terms, Conjecture 5.1 of Part I reads

$$\Delta_L(N,N) \cos^2 \angle(G_{m,m'}, P_N) \;\ll_\varepsilon\; \frac{N^\varepsilon}{\|G_{m,m'}\|^2} \qquad \big((N, mm') = 1,\ v_2(N) \leq 1\big), \tag{10}$$

and since $\|G\|^2 \ll (mm')^{1/2+\varepsilon} \ll L^{1/2+\varepsilon}$, the right-hand side of (10) exceeds the average (9) by a factor of at least $L^{1/2-\varepsilon}$. Thus the conjecture does not ask for square-root cancellation at every N, only that no N be aligned with G more than $L^{1/2}$ times the average; what is missing is a bound for the alignment that holds for every N rather than on average.

What such a bound requires. Expanding both sides in an orthogonal basis $\mathcal{B}^*$ of Hecke eigenforms as in Part I, Section 5.5,

$$\langle G, P_N \rangle = \frac{1}{4\pi N} \sum_{\varphi \in \mathcal{B}^*} \frac{\langle \Theta, \varphi \rangle\, \overline{a_\varphi(N)}}{\|\varphi\|^2},$$

the Cauchy–Schwarz bound $|\langle G, P_N \rangle| \leq \|G\| \|P_N\|$ is attained only if the vector $(\langle \Theta, \varphi \rangle / \|\varphi\|)_\varphi$ is proportional to $(a_\varphi(N)/\|\varphi\|)_\varphi$. The missing lemma is therefore a statement about the *correlation*

between the two families $\langle \Theta_{m,m'}, \varphi\rangle$ (fixed by m, m', independent of N) and $a_\varphi(N)$ (independent of m, m' beyond the level), as φ runs over $\mathcal{B}^*$:

$$\Big|\sum_{\varphi\in\mathcal{B}^*} \frac{\langle \Theta, \varphi\rangle\,\overline{a_\varphi(N)}}{\|\varphi\|^2}\Big| \ll_\varepsilon N^{1/2+\varepsilon} \qquad \text{uniformly in } m, m' \text{ and } N \text{ with } (N, mm') = 1. \tag{11}$$

Theorem 1.1 shows that (11) holds in mean square over N with the stronger right-hand side $N^{1/2}(mm')^{-1/4+\varepsilon}$. We do not know how to prove (11) for individual N; Part I, Section 5.5, discusses the approximate-functional-equation approach and its obstacle.

5. Numerical evidence

The script `mean_square_check.py` computes, with the tables and the FFT convolution of `sup_error_scan.py` of Part I, the sums $S_2(Y) = \sum_{N\in\mathcal{N}_Y} |a_G(N)|^2$ over the admissible set

$$\mathcal{N}_Y = \{N \le Y : \ (N, mm') = 1, \ v_2(N) \le 3\},$$

on which the main term of that script is exact, for $Y = 2^{21}$ and $Y = 2^{20}$, and the estimate

$$\|G\|^2_{\mathrm{est}} = \frac{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})}{16\pi^2} \cdot \frac{1}{\#\mathcal{N}_Y} \sum_{N\in\mathcal{N}_Y} \frac{|a_G(N)|^2}{N}$$

suggested by the Rankin–Selberg asymptotic of Remark 2.1 (it assumes that the mean square per N is the same on the excluded N, which is immaterial when mm' has only large prime factors). Table 1 shows the result.

(m, m')	mm'	$\#N$	$S_2(Y)/S_2(Y/2)$	$\|G\|^2_{\mathrm{est}}$	$\|G\|^2_{\mathrm{est}}/(mm')^{1/2}$	$\|G\|^2_{\mathrm{est}}/(mm')^{1/4}$
$(1, 5)$	5	1572864	4.001	0.74	0.33	0.50
$(9, 1)$	9	1310720	4.001	1.47	0.49	0.85
$(3, 5)$	15	1048576	4.001	3.44	0.89	1.75
$(1, 101)$	101	1946614	4.000	7.95	0.79	2.51
$(13, 29)$	377	1752262	4.001	13.09	0.67	2.97
$(1, 1009)$	1009	1964131	3.976	17.96	0.57	3.19
$(61, 67)$	4087	1904986	3.961	24.47	0.38	3.06
$(101, 103)$	10403	1927715	3.948	29.81	0.29	2.95
$(1, 15015)$	15015	754220	4.359	36.62	0.30	3.31
$(1009, 1013)$	1022117	1962192	4.559	46.87	0.046	1.47

Table 1. Mean square of the cuspidal error over the admissible $N \le Y = 2^{21}$ (`results/mean_square_check_x21.txt`). The ratio $S_2(Y)/S_2(Y/2) \approx 4$ is the growth Y^2 of Theorem 1.1. For the first three pairs a large proportion of the N is excluded by the coprimality condition, and for the last pair $Y < L$, so that neither (1) in its Y^2/L form nor the Rankin–Selberg asymptotic applies; these rows are shown for completeness.

Observation 5.1. For the six pairs with $10^2 \le mm' \le 1.6\cdot 10^4$ and $Y \ge 35L$, the estimate $\|G_{m,m'}\|^2_{\mathrm{est}}$ lies between $2.5\,(mm')^{1/4}$ and $3.3\,(mm')^{1/4}$, while $\|G\|^2_{\mathrm{est}}/(mm')^{1/2}$ decreases from 0.79 to 0.29 over the same range. The data therefore suggest

$$\|G_{m,m'}\|^2 \asymp (mm')^{1/4}, \qquad \text{i.e.} \qquad \|G_{m,m'}\| \asymp (mm')^{1/8},$$

rather than the bound $(mm')^{1/4+\varepsilon}$ of Lemma 2.1 of Part I. This corrects the remark in Part I, Section 5.3, that the root mean square "decreases like $(mm')^{-1/4}$" and that Waibel's bound "appears to be sharp": over the reliable range the root mean square of $(T-M)/\sqrt{N}$ decreases like

$(mm')^{-3/8}$, and the second term $N^{1+\varepsilon}/(\det A)^{1/4}$ of Waibel's Theorem 1 appears to overestimate $\|G_{m,m'}\|^2$ by a factor of about $(mm')^{1/4}$ for this family.

We record what a norm bound of this strength would give, as a conditional statement.

Hypothesis 5.2. $\|G_{m,m'}\| \ll_\varepsilon (mm')^{1/8+\varepsilon}$ for all coprime odd m, m'.

Proposition 5.3. *Under Hypothesis 5.2: Theorem 2.3 and Corollary 2.7 of Part I hold with $(mm')^{1/8+\varepsilon}$ in place of $(mm')^{1/4+\varepsilon}$; the level of distribution of $r(n)r(N-n)$ is $4/9$ for every $N \equiv 2 \pmod 4$ and $4/5$ for almost all such N; and the sieve of Part I gives $\Omega(n(N-n)) \le 8$ and $\max(\Omega(n), \Omega(N-n)) \le 6$ for every N.*

Proof. The proofs of Theorem 2.3 and Corollary 2.7 of Part I use the norm only through Lemma 2.1. In Theorem 3.2 of Part I the exponent $1/4$ enters through $\sum_{k \le D} d(k) k^{1/4} \ll D^{5/4} N^\eta$; with $1/8$ the balance $N^{1/2} D^{9/8} = N$ gives $D = N^{4/9}$ (Remark 3.3 of Part I), and in Theorem 1.2 the balance $X^{3/2} D^{5/8} = X^2$ gives $D = X^{4/5}$. The floating-point scan at $\theta = 4/9$ gives, at best, $1/\lambda + 2/\beta_1 \approx 8.61$ and $1/\lambda + 1/\beta_1 \approx 5.98$. The second value, however, is attained only at the largest admissible λ, where $L - \lambda R = 0$; any λ with a positive margin $L - \lambda R > 0$ pushes $1/\lambda + 1/\beta_1$ above 6. Hence $(8, 6)$ is attainable with margin and can be certified as in Part I, while $(8, 5)$ is not reachable with these weights at this level; we claim only $(8, 6)$. □

Hypothesis 5.2 concerns the norm alone and is a statement about a single theta series, not about its individual coefficients; it might be more accessible than Conjecture 5.1 of Part I. Waibel's lower bound (Theorem 3 of [3]) gives only $\|G_{m,m'}\|^2 \gg 1/\min(m, m') + O(L^\varepsilon)$ for these forms and is far from the truth here; the growth $(mm')^{1/4}$ of $\|G\|^2$, if real, asks for an explanation in terms of the spectral weights $\langle \Theta, \varphi \rangle$. By Parseval over the orthogonal eigenbasis $\mathcal{B}^*$,

$$\|G\|^2 = \sum_{\varphi \in \mathcal{B}^*} \frac{|\langle \Theta, \varphi \rangle|^2}{\|\varphi\|^2},$$

so Hypothesis 5.2 is the statement that this weighted second moment over the family of $\asymp L$ eigenforms is $\ll L^{1/4+\varepsilon}$. In the heuristic factorisation of Part I, Section 5.5, $\langle \Theta, \varphi \rangle = \mathcal{E}_\varphi(m, m') L(\frac{1}{2}, f) L(\frac{1}{2}, f \otimes \chi)$, and $\|G\|^2$ becomes a second moment of products of two central values with the weights $|\mathcal{E}_\varphi|^2/\|\varphi\|^2$. (Since θ^2 is an Eisenstein series, the inner product is a Rankin–Selberg integral, not a triple product of cusp forms.) Without the local factors $\mathcal{E}_\varphi$, which have not been worked out, this does not predict the exponent $1/4$; it does locate the question in the second moment of a family of L-values, where tools exist.

Data and code

The scripts `mean_square_check.py` and `certify_modular.py` (argument `2/3`), the modules of Part I they use (`sup_error_scan.py`, `modular_sieve_scan.py`, `sieve_core.py`) and the outputs `results/mean_square_check_x21.txt`, `results/certify_modular_output_twothirds.txt` are in the repository https://github.com/Ruqing1963/two-squares-mean-square; the code and data of Part I are at https://github.com/Ruqing1963/two-squares-theta-level. The certification of the sieve parameters uses directed rounding as described in Part I, Appendix A; the mean-square computation is in floating point and is evidence, not proof.

References

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GUT Geoservice Inc., Montreal, Canada
Email address: ruqing@hotmail.com